

UNIVERSITY OF EDINBURGH
COLLEGE OF SCIENCE AND ENGINEERING
SCHOOL OF INFORMATICS

COMPUTER COMMUNICATIONS AND NETWORKS

Monday 11 May 2009

14:30 to 16:30

Year 3 Courses

Convener: K Kalorkoti

External Examiners: A Frisch, J Gurd

INSTRUCTIONS TO CANDIDATES

Answer any TWO questions.

All questions carry equal weight.

CALCULATORS MAY BE USED IN THIS EXAMINATION

1. (a) Describe the service model of the following: TCP, UDP, IP and Ethernet. [4 marks]
- (b) A high-quality version of an hour long TV programme on BBC iPlayer has a size of 350MB to download to a home computer. A home user has a broadband service with a download speed of 2Mbps. Round-trip time to the BBC iPlayer website is found to be 100ms using a ping test with default packet size of 56bytes. Assume that the access link is the bottleneck and delays other than due to propagation and transmission to be negligible.
 - Calculate the time it takes for the user to download the whole programme and the throughput of the transfer. [4 marks]
 - How much does the propagation delay matter for this transfer? [2 marks]
- (c) Describe mechanisms for transition from IPv4 to IPv6. [4 marks]
- (d) Give two reasons why networks might use an error-correcting code instead of error detection and retransmission. [4 marks]
- (e) Suppose your organization decides to upgrade the existing Ethernet from 10 Mbps to 100 Mbps without changing the network configuration, node locations and cabling.
 - What would you as a network designer do to maintain the same efficiency as before? [3 marks]
 - Suppose that you are also tasked with improving the efficiency of the network after the upgrade. What would you recommend your users to do? Discuss your approach. [4 marks]

2. (a) Address Resolution Protocol (ARP) and Domain Name System (DNS) both depend on caches; ARP cache entry lifetimes are typically 10 minutes, while DNS cache lifetimes are on the order of days. Justify this difference. What undesirable consequences might there be in having too long a DNS cache entry lifetime? [6 marks]
- (b) Describe three ways of searching for information in a peer-to-peer (P2P) network. [3 marks]
- (c) What does pipelining mean in the context of reliable data transfer protocols? Name two well-known pipelined reliable data transfer protocols and mention the key difference between them. Give an example of an application layer protocol in which a pipelining-like approach is used. [3 marks]
- (d) Discuss the relationship between TCP reliable data transfer approach and well-known pipelined protocols. [3 marks]
- (e) Define multiplexing and demultiplexing at the transport layer. Discuss how demultiplexing differs between TCP and UDP sessions. [4 marks]
- (f) Consider the effect of using slow-start on a link with a 10ms round-trip time and no congestion. The receive window is 24KB and the maximum segment size is 2 KB. How long does it take before the first full window can be sent in one transmission round? [2 marks]
- (g) Suppose that the TCP congestion window is set to 16 KB and a timeout occurs. How big will the window be after the next six transmission rounds, assuming they are all successful? Assume that the maximum segment size is 1KB. [4 marks]

3. (a) How does Skype enable communication between two peers behind Network Address Translators (NATs)? For what other function does Skype employ a P2P technique? [4 marks]
- (b) What is the difference between routing and forwarding? [2 marks]
- (c) Compare and contrast link-state and distance-vector routing algorithms in terms of message complexity, convergence speed and robustness. [6 marks]
- (d) What is the counting-to-infinity problem? Describe a mechanism to alleviate it. [4 marks]
- (e) Describe the Ethernet Media Access Control (MAC) Protocol. [5 marks]
- (f) An upper-layer packet is split into 5 frames, each of which has an 80 percent chance of arriving undamaged. If no error control is done by the data link protocol, how many times must the message be sent on average to get the entire message through? [4 marks]